

Bank to Non-Bank Lending and the Reallocation of Credit
by Jian Li, Yiming Ma, Caterina Mendicino and Dominik
Supera

Discussion by Quentin Vandeweyer (London Business School)
12th Wharton Conference on Liquidity and Financial Fragility

Short summary

- ▶ **Research question:** Does bank lending to non-banks crowd out loans to firms?
- ▶ **Approach:** AnaCredit (2019–25) + a bank–NBFI portfolio model
- ▶ **Findings:**
 - Bank lending to NBFIs grew $\approx 80\%$ over 2019–25; to firms, $\approx 30\%$
 - Mostly repo to funds that hold securities, not to private credit
 - Driver: funds borrow to buy the extra bonds from QT and issuance
 - Table 6: when the NBFI share rises 1pp, the firm-loan share falls 0.56pp
 - Model: a 5% shift in bond supply $\Rightarrow$ total firm credit -6.3%
- ▶ **Convincing facts.** My discussion: how to interpret them, and the mechanisms

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Short summary: the paper's mechanism

Central Bank

Bonds 9	Reserves 9
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Bank

Reserves 9	Deposits 18
Repo 2	
Loans 10	Equity 3

Pensions & insurers (LDI)

Bonds 4	Liabilities 6
Deposits 2	
long futures 2	

NBFI (hedge funds)

Bonds 2	Repo 2
short futures 2	

- ▶ Banks dislike duration more than funds $\Rightarrow$ funds buy the new bonds, on repo
- ▶ Repo charged like a loan ($\approx 72\%$), capital binds $\Rightarrow$ repo crowds out firm loans
- ▶ Experiment (5% bond-supply shift): NBFI loans +3.6%, firm credit -6.3%

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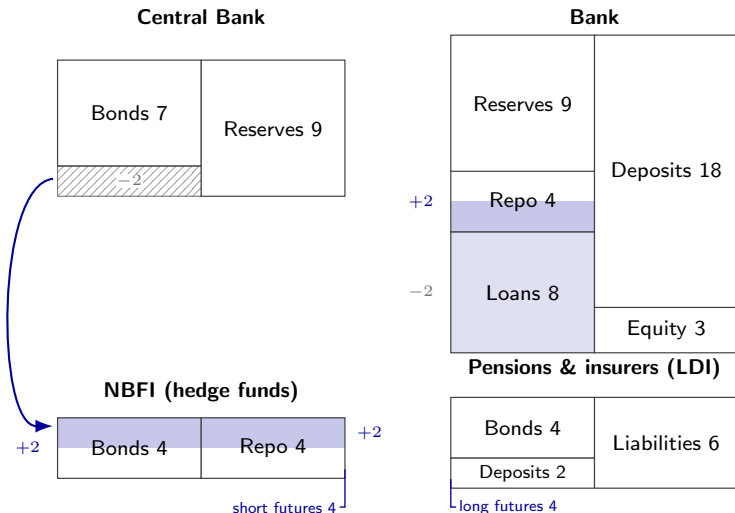
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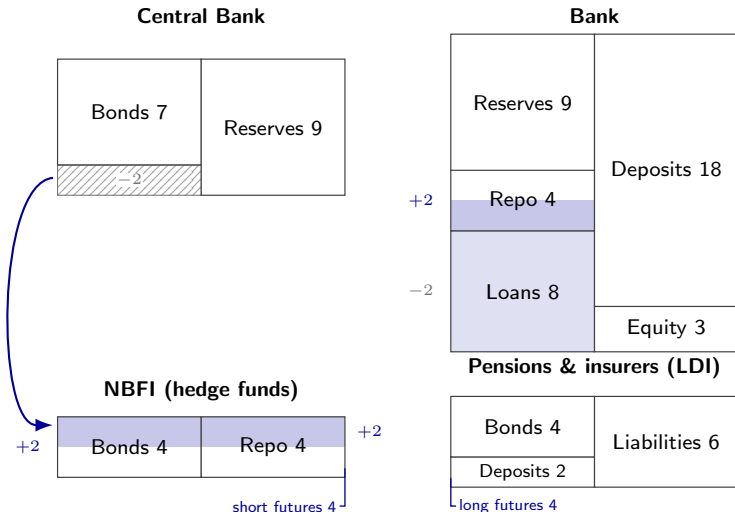
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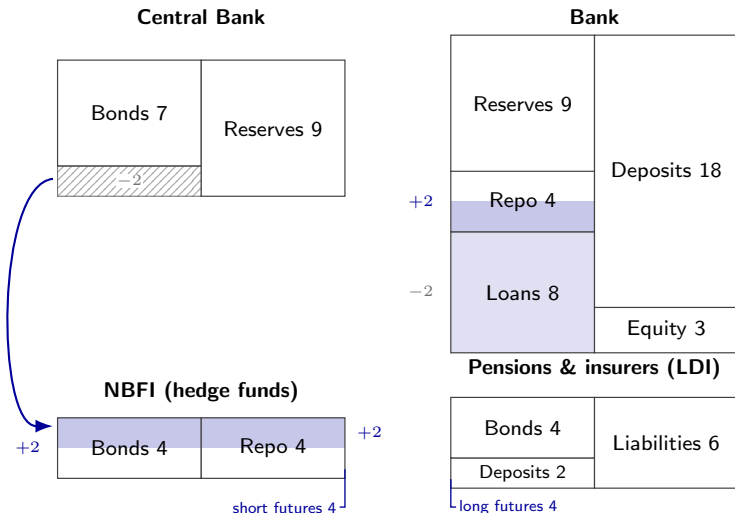
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Plan for the discussion

- ① QT drains reserves too: repo can replace them without squeezing firm loans
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1a. Well-collateralised repo uses almost no risk-weighted capital

Equiv. risk weight	Paper's model	Basel rules
Reserves	0%	0%
Government-bond repo	$\approx 72\%^*$, pooled NBFI	$\approx 0\%$ if the bonds cover it

*My calculations: the paper's A5 capital formula at its pooled NBFI-loan inputs

- ▷ **At the paper's weights**, repo uses scarce capital and can squeeze firm loans
- ▷ Paper's p.18: such repos "have a risk weight of zero"
- ▷ **Why it matters**: repo drives growth; pooled weights can overcharge it
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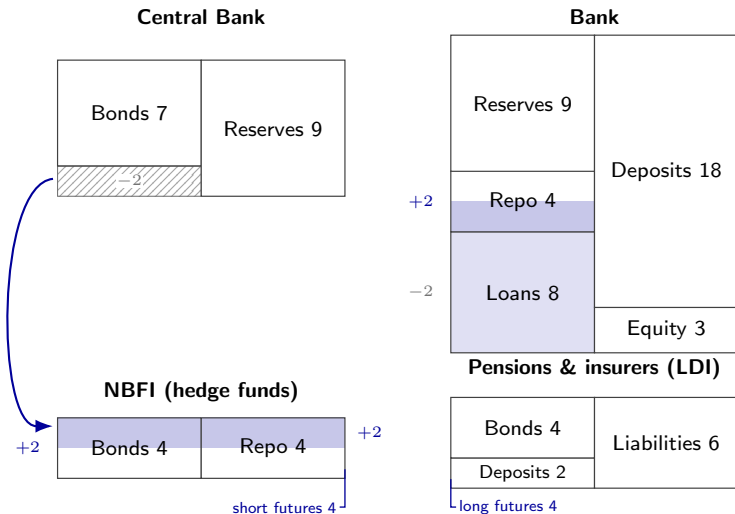
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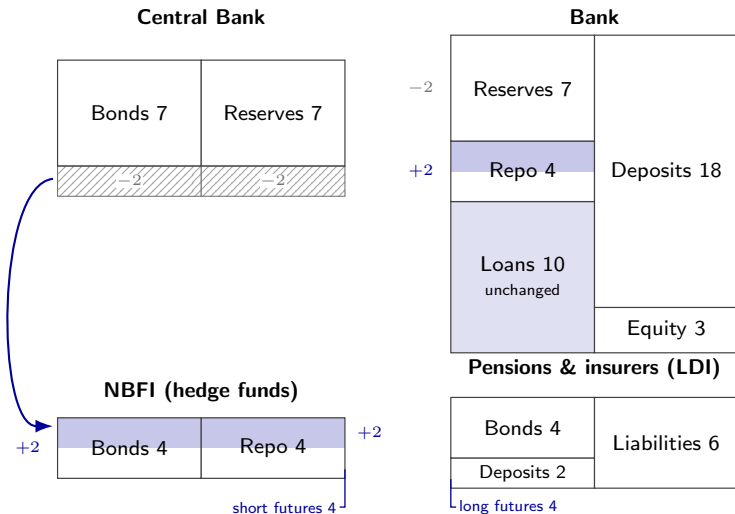
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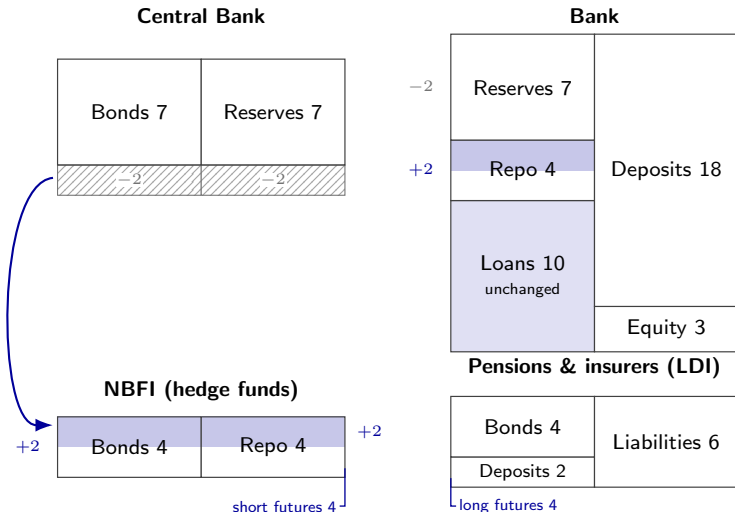
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1c. Table 6 is estimated in shares, read in euros

Estimated (eq. 4.6)

$$\Delta\left(\frac{\text{Firm loans}}{\text{Loans}+\text{securities}}\right) = \beta \cdot \Delta\left(\frac{\text{NBFI loans}}{\text{Loans}+\text{securities}}\right)$$
$$\beta = -0.56$$

Read as a level change, not a proportion change:

"... associated with a 0.56 percentage point decrease in firm loans" (p.19)

"the reallocation of credit away from firm lending" (p.19)

"largely at the expense of loans to non-financial firms" (p.3)

- ▷ **Why it matters:** if reserves aren't counted in loans + securities, swapping them for repo makes that total grow, so the firm-loan ratio falls while firm loans stay the same

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1d. How much? A pure swap gives about a third of the -0.56

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- ▶ A pure reserves-for-repo swap then gives $-c/(1 - n) = -0.17/0.90 \approx -0.19$
- ▶ So about a third of the paper's -0.56 can arise with no firm loan lost
→ Ask: Table 6 in euros on a fixed base; reserves and total book as outcomes

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2a. When buyers pay out of deposits, bank liquidity falls

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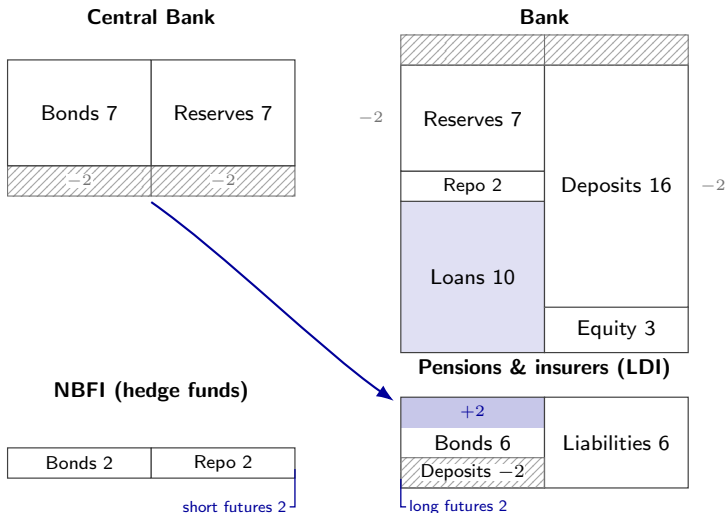
Liquidity coverage ratio: reserves $\geq \frac{1}{2}$ deposits: $9 = 9 \checkmark$

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► **Liquidity:** reserves count in full, lost deposits partly; the ratio can fall

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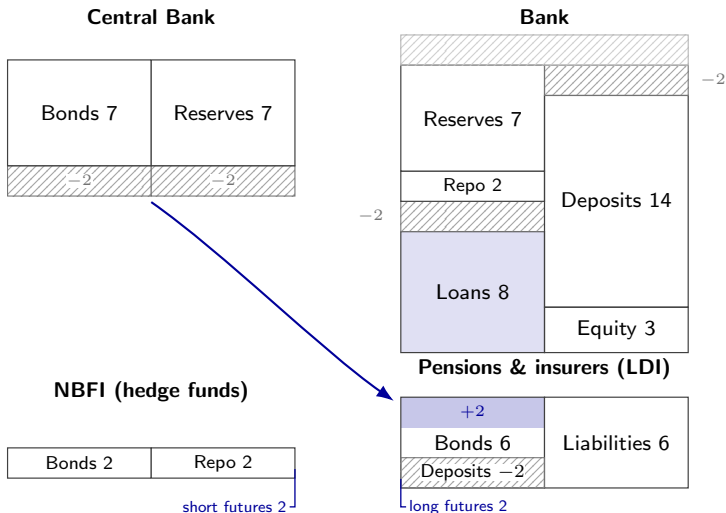
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2b. Both channels can be at work: model both, then split the effect

- ▶ **Paper's channel:** funds' demand for repo uses bank capital; firm loans give way
- ▶ **Liquidity channel:** reserves and deposits leave; banks short of liquidity lend less
- ▶ **Both fit the facts:** more repo and fewer firm loans at the same banks
- ▶ **Paper's model** has no liquidity constraint, so it cannot split the two
 - Ask: add a liquidity constraint; how much of the firm-credit fall is each?
 - Test: In Table 7, add banks' pre-QT liquidity buffer next to NBFI share

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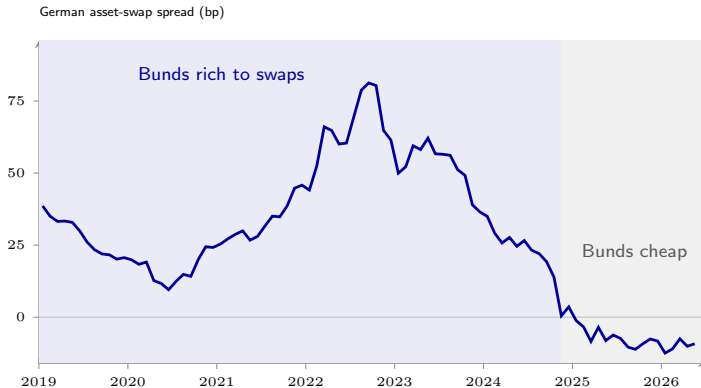
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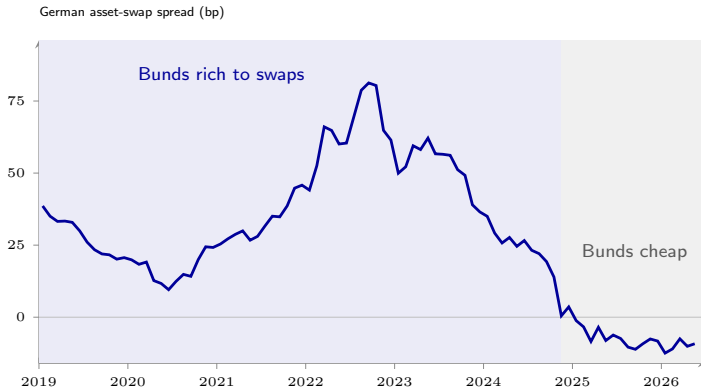
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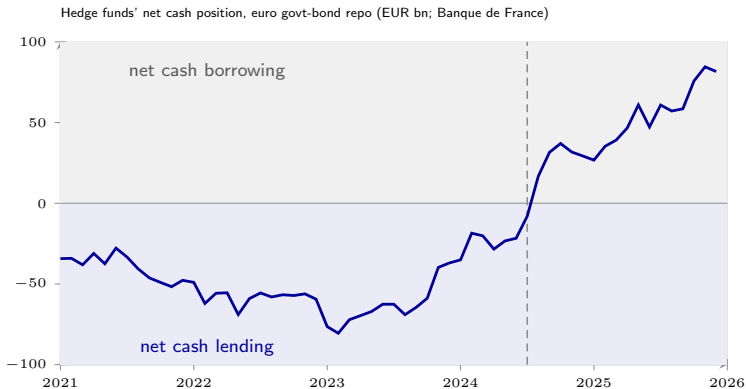
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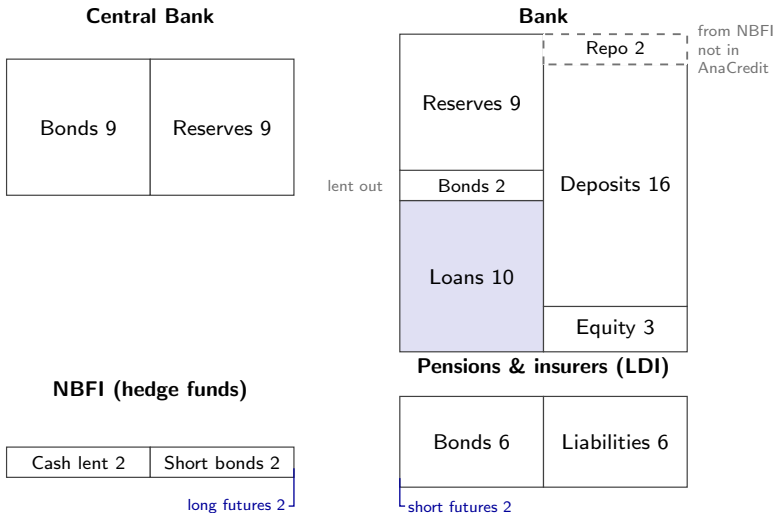
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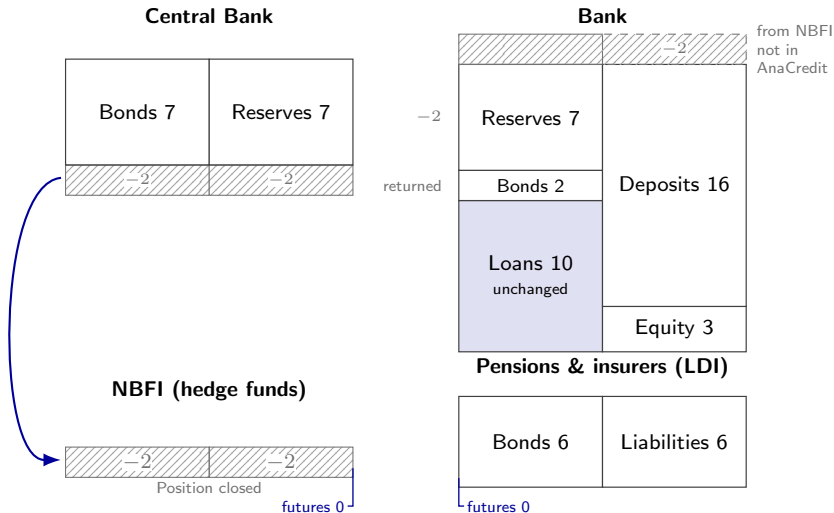
- ▷ Net cash lenders until mid-2024; net borrowers since

3c. Until mid-2024, the funds held the opposite position



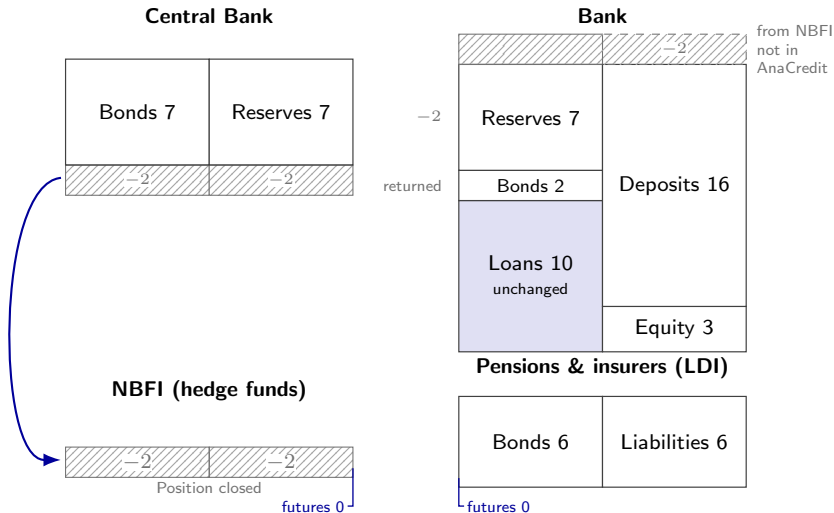
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- ▶ **Illustration:** short Bunds, lend cash, long futures; pensions hold bonds
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3d. In dollars, funds borrow bank cash, as in the paper's story

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Basis	Bunds rich until 2025	usual: long cash, short futures
Hedge funds' net cash	lent until mid-2024	borrowed, $\approx$ \$1.8tn in 2025

- ▷ **Paper's fn. 8:** a marked rise in non-euro repo, "Treasury basis trades"
- ▷ **ECB:** euro-area banks intermediate it via US affiliates, on balance sheet
- ▷ **Open question:** does this balance-sheet use crowd out euro-area firm loans?
→ Ask: split repo by currency; relate dollar books to leverage headroom

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Taking stock

- ▶ **The facts are convincing:** banks lend more to NBFIs, mostly in repo
- ▶ What drives them is still open
 - Repo can replace reserves at almost no capital cost
 - QT also drains deposits, and banks short of liquidity lend less
 - Until mid-2024, funds lent cash in euro repo rather than borrowed it
- ▶ A more comprehensive model
 - Repo priced by the rules, and reserves that fall with QT
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Appendix: what the rules charge on repo

Rule	What the bank is charged on
Bilateral: comprehensive	Exposure net of haircut-adjusted collateral; capital on the residual.
Supervisory haircuts	Default ten-day haircuts start at 0.5% for short, high-grade sovereigns; scale for repo holding period and margining.
Eligible core participants	Zero haircut: 0%-RW sovereign, same currency, overnight or daily margin.
Qualifying CCP	Trade exposure carries a 2% risk weight.
Leverage ratio	Securities financing counts gross, with limited netting.
Liquidity ratio	Available, retained collateral may count as HQLA; Level 1 has 0% maturity inflow.

- ▷ **Risk-weighted capital:** ≈ 0 when haircuts cover exposure.
 - **Caveat:** Residual exposure remains if collateral or haircuts are insufficient.

Appendix: the paper's facts in the inverted regime

Paper's finding	Paper's reading	Our reading to test
Repo to hedge funds grows	Funds need bank cash to hold bonds	Closing shorts can shrink bank repo borrowing, unseen in AnaCredit
Repo rates rise with volume	Demand for bank cash rises	Rates rose from ≈ 40 bp below the DFR to about the DFR
ECB-held share falls: more repo, higher rates	More financing needed for that bond	Scarcity relief also raises repo rates
More NBFI lending, fewer firm loans	Bank cash diverted from firms	EGB repo: net cash lenders before mid-2024; split the estimates
Model: long funds absorb bonds	Banks move duration to funds	Short, hedged cash lenders lie outside this mechanism

- ▶ **Discriminating evidence:** repo-rate levels, net positions, swap spreads.
- ▶ **Our extension:** basis moves toward zero; net bank repo lending flips sign.

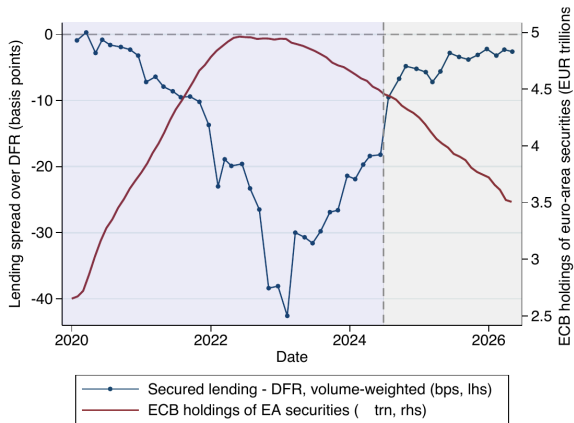
Appendix: same QT, another branch

Margin	Chavaz–Patozi–Wazzi	Li–Ma–Mendicino–Supera	Our joint reading
Lending	Bank-panel lending falls	Firm lending falls at NBFI-exposed banks	Agree on loan cuts
Exposed banks	Low liquidity; reserve-access frictions	Low reserves, deposit losers, low capital	Liquidity overlap; capital: the paper's channel only
Asset mix	Fewer loans, more liquid securities	More repo; bond holdings +€494bn	Holdings alone do not measure duration
Maturity	Average loan and asset maturity rise	Model: banks shun duration	A test of the premise; net of swaps
Exposure	Reserve-loss instrument	Initial NBFI lending share	Not tested against each other
Constraint	Rebuild liquidity; reserves leverage-exempt	Risk-weighted capital binds in model	Different mechanisms; final LCR response insignificant
Prices	Short fixes react more	Repo rates rise with volume	Repo levels also fit scarcity relief

▷ **Our reading:** liquidity loss can produce repo up and loans down.

→ **Ask:** horse race reserve losses and liquidity against NBFI exposure.

Appendix: the paper's Figure A.10



Below the DFR throughout QT

Appendix: excess liquidity fell mainly through TLTRO

- ▷ **Peak excess liquidity:** €4,748bn in November 2022.
- ▷ **Main driver through April 2024:** TLTRO III repayments
- ▷ **Also:** a gradual decline in APP holdings.
 - Ask: separate TLTRO repayments from bond runoff.

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Appendix: the dollar side

Euro-area banks	ECB evidence
Dollar repo	Almost doubled since 2022 to €1.6tn (Nov. 2024).
Collateral	70% government bonds, of which 95% Treasuries (Jan.–Nov. 2024 average).
Dollar channel	Liquidity intermediated between US affiliates and non-banks.
Net dollar borrowing	Repos exceed reverse repos by $\approx$ €250bn (Nov. 2024).
Repo intermediation	More than doubled since 2021, driven by non-EA funds.
Secured-financing books	Large intermediaries typically keep balanced dollar/euro books.

▷ **Dollars:** a separate channel through US affiliates.

→ **Ask:** separate dollar and euro repo, and identify their funding.

Appendix: aggregate timing

Period	Hedge funds, net, EGB repo	Euro-area firm credit
2023–mid-2024	net cash lenders	stalled: -0.2% (Oct 2023)
mid-2024–2025	net cash borrowers	recovering: 1.7% (Dec 2024) 2.9% (Oct 2025)

▷ **Paper's identification is cross-sectional:** aggregate timing cannot reject it

→ Ask: split Tables 6–7 at mid-2024

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Appendix: a maturity test of the two channels

	Capital: long loans → short repo (illustration)	Liquidity loss (UK evidence)
Firm loans	fall	fall
Banks add	short repo	liquid assets
Long-asset share	falls	rises

- ▶ **Paper's premise:** banks shun duration; repo for long loans shortens assets
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- ▶ **UK bank panel:** average loan and asset maturity rise
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